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# KAPASA MAKASA UNIVERSITY

## SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

### Department of Information and Communication Technology

### FINAL EXAM 2025

**Course Title: Fraud and Detection Technologies**

**Course Code: CYS 339**

**Duration: 3HRS**

**Total Marks: 60 Marks**

## General Instructions to Candidates:

1. Answer **Question 1** and **Question 2** (Compulsory) and **any two** from Questions 3, 4, and 5.
2. No unauthorised materials are allowed.
3. You are allowed to use calculators (but programmable calculators are not permitted)
4. Start each answer on a new page.
5. Write clearly and neatly.

**Note:** Any form of examination malpractice will result in disciplinary action.

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Q2 = -7000 favorable

48000 | 55000
-55000 | -48000
-7000 | 7000 (lower costs)

Q1

Variance = Actual - Expected

= 65000
- 50000
= 15000 unfavorable (higher costs)

## SECTION A (Compulsory)

### * QUESTION 1

A cybersecurity team is analysing the effectiveness of their new intrusion detection system (IDS) over two quarters.

In Quarter 1 (Q1) the budgeted security incident response cost $50,000 while the actual cost was $65,000 whilst in Quarter 2 (Q1) the budgeted security incident response cost was $55,000 with the actual cost being $48,000. The total number of alerts in Q2 was 12,000 with the number of true positive alerts (actual threats) being 960

6 12 18 24 30 36 42 48 → 8 54 60 → 10

(6) a) Calculate the cost variance for both Q1 and Q2. For each, state whether it is a favorable or unfavorable variance and explain what this means in the context of cybersecurity spending. [6 marks]

4/5 × 100%

960 : 12000
48 : 60
8 : 10
4 : 5

b) Calculate the False Positive Ratio for the IDS in Q2. [3 marks]

(6) c) Based on your calculations, advise the Chief Cybersecurity Officer on two reasons why relying solely on these financial ratios and variances might give an incomplete picture of the IDS's performance. [6 marks]

Financial
(i) oversimplification
(ii) data accuracy issues
(iii) lagging indicators

960/12000 × 100% = 8%

### * QUESTION 2 (15)

(4) a) Define Gap Analysis and explain its importance in fraud detection. [4 Marks]

(5) b) Outline the five (5) key steps in conducting a gap analysis for fraud prevention in an organisation. [5 Marks]

(6) c) A company's fraud detection system has a high false-positive rate. Using gap analysis, suggest three (3) improvements to reduce false alerts. [6 Marks]

Variance
(i) Sensitive to outliers
(ii) false positives
(iii) Low variance attacks

Define the objectives
Assess the current state
Identify the desired state
Identify the gap
Develop an action plan

(i) Compare rule-based vs. AI/ML
(ii) Data consistency
(iii) Avoid delayed responses

APPROXIMATE

Q1 = 12
Q2 = 15
Q3 = 11
Q4 = 15
53

TOTAL = 53
CA = 33
86 → A

**SECTION B (Answer any two)** ✓

**QUESTION 3**

(6) a) Compare qualitative and quantitative data analysis in fraud detection by providing two (2) strengths and two (2) weaknesses for each. [6 Mark] ✓

(5) b) Explain how behavioral profiling can detect fraudulent transactions in banking. [5 Marks] ✓

c) A fraud analyst is investigating an unusual spike in refund requests. Recommend two quantitative and two qualitative techniques to determine if this is fraud. [4 Marks] ✓

keywords
context
Behavioral

SDP² CT
Predictive and Behavioral
Time Series Contexual

**QUESTION 4** ✓

(4) a) Briefly explain the four (4) stages of the OLAF Approach in fraud investigations. [4 Mark] ✓

object
locate
analyze
Follow

15

(5) b) How can link analysis (under the "Follow" stage) help uncover a vendor fraud scheme? [5 Marks] ✓

(6) c) Discuss two (2) challenges investigators may face when applying the OLAF approach in digital forensics. [6 Marks] ✓

Encryption and privacy concerns
data overload slows analysis

**QUESTION 5**

a) Explain three (3) considerations when selecting between generic and specific fraud detection tools. [6 Marks]

b) Describe how critical thinking and logical reasoning help in formulating fraud hypotheses. [4 Marks]

c) A company suspects payroll fraud. Outline a step-by-step data analysis plan to investigate. [5 Marks]

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